

Carnegie Foundation Practice

Name: _____ Period: _____ Date: _____

STANDARD	LEARNING INTENTION	SUCCESS EVIDENCE
6.RP.1	Compare two quantities and build the meaning of a ratio.	Name the quantities, preserve order, label the ratio, and identify the whole.

Directions: Complete all eight problems on paper. Label each quantity. Do not simplify away the meaning of a ratio. Explain your reasoning when asked.

1. Multiplicative comparison

Juan has 7 apples. Latanya has 28 apples, or 4 times as many as Juan. Is this additive or multiplicative reasoning? Name the factor and explain how you know.

2. Additive comparison

Melanie is 8 years old. Justin is 4 years older than Melanie. Is this additive or multiplicative reasoning? Explain why the number 4 describes a difference rather than a ratio.

3. Three forms, same order

A cereal-bar box has 5 blueberry bars, 3 strawberry bars, and 2 apple bars. Write the ratio of strawberry bars to blueberry bars in words, colon notation, and fractional form with labels.

4. Find the whole

Using the same cereal-bar box, write the ratio of apple bars to the total number of bars in colon notation and fractional form with labels. Classify it as part-to-part or part-to-whole.

5. Part-to-part

A movie list has 5 sports movies, 3 science-fiction movies, and 2 comedies. Write sports movies to science-fiction movies in colon and fractional form with labels. Classify the ratio.

6. Part-to-whole

Using the same movie list, write comedies to total movies in colon and fractional form with labels. Classify the ratio and state what the whole represents.

7. Error analysis

A model has 3 blue counters and 2 yellow counters. A student writes, 'The ratio of blue counters to the whole is 3:2.' Identify the error, write the correct ratio, and explain what each number represents.

8. Transfer

Of 31 students surveyed, 19 prefer white bread. The rest prefer wheat bread. Determine how many prefer wheat bread. Then write one part-to-part ratio and one part-to-whole ratio. Label every quantity and explain why order matters.

Curriculum foundation: Adapted for classroom use from Carnegie Learning Course 1, M2.T1 Skills Practice, pages 43-45, and Lesson 1 Assignment, page 181.